



EdgeResilience

Snapdragon-Powered Predictive Intelligence for Connected Vehicle Safety

Predicting connected-vehicle resilience degradation at the edge, on Snapdragon.

The Connected Vehicle Resilience Challenge

Connected vehicles must stay safe even when the cloud is unreachable.

Threats considered: CAN bus anomalies, message injection, timing manipulation and payload tampering.

Predictive

Detect degradation before it becomes critical, not after.

Resilient

Keep local intelligence running when connectivity is lost.

Auditable

Preserve tamper-evident evidence for post-event analysis.

Local compute must carry detection, containment and evidence until connectivity returns.

EdgeResilience Solution

A complete edge AI pipeline: predict, respond locally, preserve evidence.

- Reads 17 telemetry features over 12 temporal steps
- Predicts future resilience degradation with a Temporal Transformer
- Applies a deterministic, explainable risk policy
- Keeps inference local, whatever the connectivity state
- Buffers SHA-256 hashed evidence while degraded, then prepares a synchronization manifest on recovery

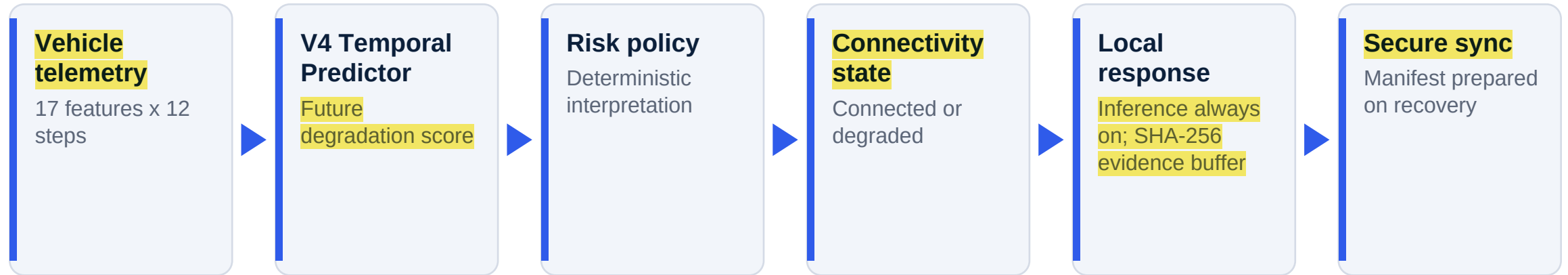
Why the edge?

Low latency, no cloud dependency, and evidence that stays local until it can be synchronized.

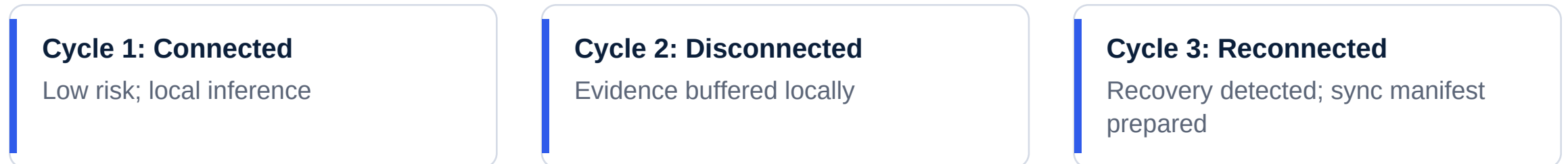
Snapdragon NPU execution makes this practical on a PC-class device.

System Architecture

From vehicle telemetry to prediction, local response and forensic evidence.



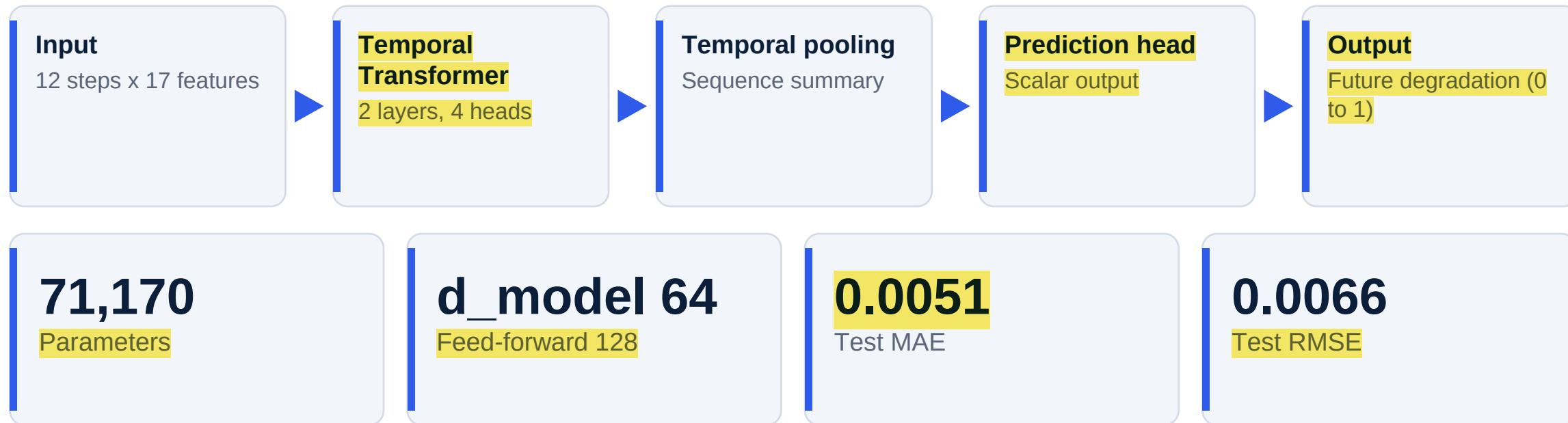
Demonstrated three-cycle scenario



Synchronization is a software manifest; no real network transmission is performed.

V4 AI Architecture

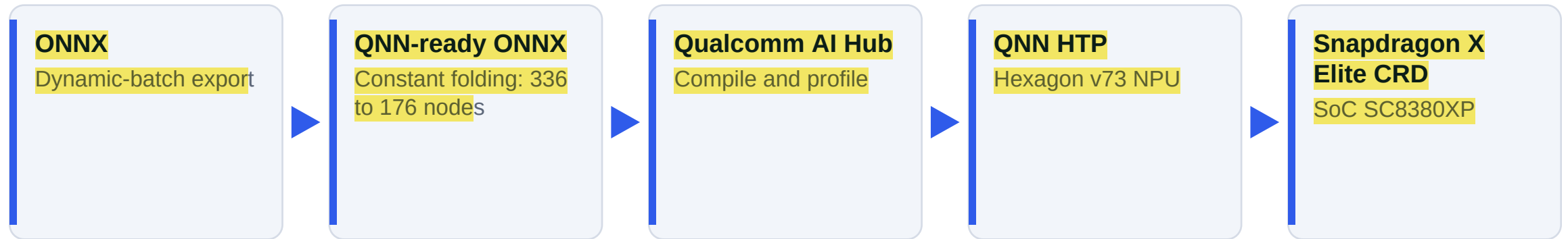
A compact Temporal Transformer that reads the full 12-step history.



- A temporal ablation showed the full 12-step sequence outperforms final-step-only and compact-summary inputs.
- Evaluated on a synthetic 5,000-record temporal dataset; not raw vehicle telemetry.

Snapdragon Deployment

The V4 workload was compiled and profiled through Qualcomm AI Hub.



- ONNX export matched PyTorch on CPU (max error 5.96e-08, threshold 1e-05).
- QNN-ready graph matched the source ONNX on CPU (max error 0.0 over 200 samples).
- Snapdragon results come from the X Elite CRD reference device.

Development machine: Lenovo IdeaPad 3 15IAU7 (Intel Core i5-1235U). Snapdragon evidence: X Elite CRD.

Validated Production Performance

Full-production V4 model, profiled on Snapdragon X Elite CRD (QNN HTP).

63 μ s

Estimated inference latency

67 μ s

Warm median

68.96 μ s

Warm mean

77 / 82 / 88 μ s

P90 / P95 / P99

28.5 MB

Peak inference memory

90 / 90

Execution entries on NPU

99 warm single-batch measurements averaged 68.96 μ s with a 67 μ s median. The warm profile-derived rate is approx. 14,501 inferences per second; it is not a separate sustained-throughput benchmark.

Workload-specific results; not full vehicle-system latency.

Numerical Validation & Evidence Integrity

An honest engineering boundary, documented and being addressed.

What is established

- Full V4 model executes on the Snapdragon NPU
- Latency and memory profiled on X Elite CRD
- CPU-side ONNX equivalence passes

Current limitation

- The CPU-to-Snapdragon comparison was executed and did not meet the $1e-05$ threshold
- Max absolute error: 0.0030923495
- Max relative error: 38.50%

We therefore make no claim of CPU-to-Snapdragon numerical equivalence. Operator-level investigation is the next engineering step.

Automotive Safety & Cybersecurity Position

Research validation is kept clearly separate from operational deployment.

Validated in this work

- Software simulation and virtual laboratory
- Snapdragon X Elite CRD profiling
- Local detection, buffering and recovery logic
- Tamper-evident (SHA-256) evidence records

Outside current scope

- Physical vehicle testing
- External CAN transmission or direct actuation
- Production safety-controller deployment
- Certified safety limits (thresholds are demonstration defaults)

Also outside scope: power and thermal measurement, HTP utilization, and validation on a physical HP Snapdragon laptop.

Innovation & Impact

One pipeline that joins prediction, local response and evidence.

Connected vehicles

Anticipates degradation instead of only reacting to it.

Edge AI

Sub-100 μ s profiled inference on a Snapdragon NPU.

Automotive cybersecurity

Local response with tamper-evident evidence records.

Resilience

Operation continues through connectivity loss and recovery.

Snapdragon computing

A documented ONNX to QNN HTP path on X Elite.

Evidence-first

Every claim here maps to a repository artifact.

Future Development

Realistic engineering steps, in priority order.

1

Resolve numerical divergence

Operator-level analysis, corrective graph changes, re-run the full comparison.

2

Extended Snapdragon benchmarking

Broader profiling; power and thermal measurement where available.

3

Real-time streaming inference

Continuous telemetry windows instead of batch scenarios.

4

Quantization and distillation

Evaluate accuracy and latency trade-offs.

5

V2X cooperative cyber resilience

Multi-vehicle simulation and geographic threat clustering.

Thank you

EdgeResilience demonstrates a validated path from a temporal AI model to Qualcomm QNN HTP execution on Snapdragon X Elite, with its remaining validation work stated openly.

Designed for Snapdragon-powered HP PCs. Validated to date on the Snapdragon X Elite CRD.

GitHub repository

github.com/Sureshdmesw/EdgeResilience